

✓ Candidate Number: 48372

This Notebook is for providing answers to Question 1 (MongoDB Part) Only

✓ Question (A)/(1A)

A.1 Load pymongo, pandas; Connect to MongoDB with pymongo

```
import pymongo
from pymongo import MongoClient
import pandas as pd

# Connect to MongoDB with the my credentials
uri = "mongodb+srv://48372:48372@cluster0.0ru7l.mongodb.net/?retryWrites=true&w=majority"
client = MongoClient(uri)

db = client['Northwind']
```

✓ A.2 Create Collections

Hint: you may encounter error when running this code cell, because if one collection is already created and its name is used, you cannot run the code to create it again with the same name.

```
db.create_collection("customers")
db.create_collection("employees")
db.create_collection("orders")
db.create_collection("products")
db.create_collection("suppliers")
db.create_collection("categories")
# Test connection
print("Collections:", db.list_collection_names())
```

🔄 Collections: ['customers', 'categories', 'employees', 'suppliers', 'orders', 'products']

✓ A.3 Loading CSV files into corresponding Database Collections in MongoDB

```
# Load 6 csv files. Path = current directory/data
customers_df = pd.read_csv("./data/customers.csv")
employees_df = pd.read_csv("./data/employees.csv")
suppliers_df = pd.read_csv("./data/suppliers.csv")
orders_df = pd.read_csv("./data/orders.csv")
products_df = pd.read_csv("./data/products.csv")
categories_df = pd.read_csv("./data/categories.csv")

# Data read by pandas is in the form of DataFrame, I need to convert it to dictionary for MongoDB.
customers_data = customers_df.to_dict(orient="records")
employees_data = employees_df.to_dict(orient="records")
suppliers_data = suppliers_df.to_dict(orient="records")
orders_data = orders_df.to_dict(orient="records")
products_data = products_df.to_dict(orient="records")
categories_data = categories_df.to_dict(orient="records")
```

```
# Insert data into collections
db["customers"].insert_many(customers_data)
db["employees"].insert_many(employees_data)
db["suppliers"].insert_many(suppliers_data)
db["orders"].insert_many(orders_data)
db["products"].insert_many(products_data)
db["categories"].insert_many(categories_data)
```

🔄 InsertManyResult([ObjectId('675ee815ae2ae6f2bc92f992'), ObjectId('675ee815ae2ae6f2bc92f993'), ObjectId('675ee815ae2ae6f2bc92f994'), ObjectId('675ee815ae2ae6f2bc92f995'), ObjectId('675ee815ae2ae6f2bc92f996'), ObjectId('675ee815ae2ae6f2bc92f997'), ObjectId('675ee815ae2ae6f2bc92f998'), ObjectId('675ee815ae2ae6f2bc92f999')], acknowledged=True)

✓ A.4 Deploying relationships between collections

A.4.1 From the provided Image, it could be observed that the PK(primary key) of Suppliers is the FK(foreign key) of Products. To represent

- ✓ this relationship with MongoDB, I create a new Attribute called **created_supplier_id**, which stored the information of collection Suppliers's **Object IDs**.

```
for product in db["products"].find():
    created_supplier_id = product.get("SupplierID")

    supplier = db["suppliers"].find_one({"SupplierID": created_supplier_id})

    db["products"].update_one(
        {"_id": product["_id"]},
        {"$set": {"created_supplier_id": supplier["_id"]}},
        upsert=False
    )

print(f"Updated product {product['ProductName']} with new attribute created_supplier_id {supplier['_id']}")
```

```
Updated product Chai with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c4
Updated product Chang with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0bd
Updated product Aniseed Syrup with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0bd
Updated product Chef Anton's Cajun Seasoning with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0be
Updated product Chef Anton's Gumbo Mix with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0be
Updated product Grandma's Boysenberry Spread with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0bf
Updated product Uncle Bob's Organic Dried Pears with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0bf
Updated product Northwoods Cranberry Sauce with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0bf
Updated product Mishi Kobe Niku with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c0
Updated product Ikura with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c0
Updated product Queso Cabrales with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c1
Updated product Queso Manchego La Pastora with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c1
Updated product Konbu with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c2
Updated product Tofu with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c2
Updated product Genen Shouyu with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c2
Updated product Pavlova with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c3
Updated product Alice Mutton with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c3
Updated product Carnarvon Tigers with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c3
Updated product Teatime Chocolate Biscuits with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c4
Updated product Sir Rodney's Marmalade with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c4
Updated product Sir Rodney's Scones with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c4
Updated product Gustaf's Knäckebröd with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c5
Updated product Tunnbröd with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c5
Updated product Guaránó Fantástica with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c6
Updated product NuNuCa Nuß-Nougat-Creme with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c7
Updated product Gumbär Gummiwürchen with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c7
Updated product Schoggi Schokolade with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c7
Updated product Rössle Sauerkraut with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c8
Updated product Thüringer Rostbratwurst with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c8
Updated product Nord-Ost Matjeshering with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0c9
Updated product Gorgonzola Telino with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0ca
Updated product Mascarpone Fabioli with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0ca
Updated product Geitost with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0cb
Updated product Sasquatch Ale with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0cc
Updated product Steeleye Stout with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0cc
Updated product Inlagd Sill with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0cd
Updated product Gravad lax with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0cd
Updated product Côte de Blaye with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0ce
Updated product Chartreuse verte with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0ce
Updated product Boston Crab Meat with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0cf
Updated product Jack's New England Clam Chowder with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0cf
Updated product Singaporean Hokkien Fried Mee with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d0
Updated product Ipoh Coffee with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d0
Updated product Gula Malacca with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d0
Updated product Rogede sild with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d1
Updated product Spegesild with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d1
Updated product Zaanse koeken with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d2
Updated product Chocolade with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d2
Updated product Maxilaku with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d3
Updated product Valkoinen suklaa with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d3
Updated product Manjimup Dried Apples with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d4
Updated product Filo Mix with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d4
Updated product Perth Pasties with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d4
Updated product Tourtière with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d5
Updated product Pâté chinois with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d5
Updated product Gnocchi di nonna Alice with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d6
Updated product Ravioli Angelo with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d6
Updated product Escargots de Bourgogne with new attribute created_supplier_id 675ee810ae2ae6f2bc92f0d7
```

- ✓ A.4.2 From the image, the PK ProductID of products is the FK of the order details. But because order details and orders merge into a single collection to orders. I will actually build the relationship between products and orders following the same logic as the previous one.

```
for orders in db["orders"].find():
    created_product_id = orders.get("ProductID")

    products = db["products"].find_one({"ProductID": created_product_id})

    db["orders"].update_one(
        {"_id": orders["_id"]},
        {"$set": {"created_product_id": products["_id"]}},
        upsert=False
    )

    print(f"Updated order with ID {orders['OrderID']} with new attribute created_product_id {products['_id']}")
```

```
➡ Updated order with ID 10248 with new attribute created_product_id 675ee814ae2ae6f2bc92f94f
Updated order with ID 10248 with new attribute created_product_id 675ee814ae2ae6f2bc92f96e
Updated order with ID 10248 with new attribute created_product_id 675ee814ae2ae6f2bc92f98c
Updated order with ID 10249 with new attribute created_product_id 675ee814ae2ae6f2bc92f952
Updated order with ID 10249 with new attribute created_product_id 675ee814ae2ae6f2bc92f977
Updated order with ID 10250 with new attribute created_product_id 675ee814ae2ae6f2bc92f96d
Updated order with ID 10250 with new attribute created_product_id 675ee814ae2ae6f2bc92f977
Updated order with ID 10250 with new attribute created_product_id 675ee814ae2ae6f2bc92f985
Updated order with ID 10251 with new attribute created_product_id 675ee814ae2ae6f2bc92f95a
Updated order with ID 10251 with new attribute created_product_id 675ee814ae2ae6f2bc92f97d
Updated order with ID 10251 with new attribute created_product_id 675ee814ae2ae6f2bc92f985
Updated order with ID 10252 with new attribute created_product_id 675ee814ae2ae6f2bc92f958
Updated order with ID 10252 with new attribute created_product_id 675ee814ae2ae6f2bc92f965
Updated order with ID 10252 with new attribute created_product_id 675ee814ae2ae6f2bc92f980
Updated order with ID 10253 with new attribute created_product_id 675ee814ae2ae6f2bc92f963
Updated order with ID 10253 with new attribute created_product_id 675ee814ae2ae6f2bc92f96b
Updated order with ID 10253 with new attribute created_product_id 675ee814ae2ae6f2bc92f975
Updated order with ID 10254 with new attribute created_product_id 675ee814ae2ae6f2bc92f95c
Updated order with ID 10254 with new attribute created_product_id 675ee814ae2ae6f2bc92f97b
Updated order with ID 10254 with new attribute created_product_id 675ee814ae2ae6f2bc92f98e
Updated order with ID 10255 with new attribute created_product_id 675ee814ae2ae6f2bc92f946
Updated order with ID 10255 with new attribute created_product_id 675ee814ae2ae6f2bc92f954
Updated order with ID 10255 with new attribute created_product_id 675ee814ae2ae6f2bc92f968
Updated order with ID 10255 with new attribute created_product_id 675ee814ae2ae6f2bc92f97f
Updated order with ID 10256 with new attribute created_product_id 675ee814ae2ae6f2bc92f979
Updated order with ID 10256 with new attribute created_product_id 675ee814ae2ae6f2bc92f991
Updated order with ID 10257 with new attribute created_product_id 675ee814ae2ae6f2bc92f95f
Updated order with ID 10257 with new attribute created_product_id 675ee814ae2ae6f2bc92f96b
Updated order with ID 10257 with new attribute created_product_id 675ee814ae2ae6f2bc92f991
Updated order with ID 10258 with new attribute created_product_id 675ee814ae2ae6f2bc92f946
Updated order with ID 10258 with new attribute created_product_id 675ee814ae2ae6f2bc92f949
Updated order with ID 10258 with new attribute created_product_id 675ee814ae2ae6f2bc92f964
Updated order with ID 10259 with new attribute created_product_id 675ee814ae2ae6f2bc92f959
Updated order with ID 10259 with new attribute created_product_id 675ee814ae2ae6f2bc92f969
Updated order with ID 10260 with new attribute created_product_id 675ee814ae2ae6f2bc92f96d
Updated order with ID 10260 with new attribute created_product_id 675ee814ae2ae6f2bc92f97d
Updated order with ID 10260 with new attribute created_product_id 675ee814ae2ae6f2bc92f982
Updated order with ID 10260 with new attribute created_product_id 675ee814ae2ae6f2bc92f98a
Updated order with ID 10261 with new attribute created_product_id 675ee814ae2ae6f2bc92f959
Updated order with ID 10261 with new attribute created_product_id 675ee814ae2ae6f2bc92f967
Updated order with ID 10262 with new attribute created_product_id 675ee814ae2ae6f2bc92f949
Updated order with ID 10262 with new attribute created_product_id 675ee814ae2ae6f2bc92f94b
Updated order with ID 10262 with new attribute created_product_id 675ee814ae2ae6f2bc92f97c
Updated order with ID 10263 with new attribute created_product_id 675ee814ae2ae6f2bc92f954
Updated order with ID 10263 with new attribute created_product_id 675ee814ae2ae6f2bc92f95c
Updated order with ID 10263 with new attribute created_product_id 675ee814ae2ae6f2bc92f962
Updated order with ID 10263 with new attribute created_product_id 675ee814ae2ae6f2bc92f98e
Updated order with ID 10264 with new attribute created_product_id 675ee814ae2ae6f2bc92f946
Updated order with ID 10264 with new attribute created_product_id 675ee814ae2ae6f2bc92f96d
Updated order with ID 10265 with new attribute created_product_id 675ee814ae2ae6f2bc92f955
Updated order with ID 10265 with new attribute created_product_id 675ee814ae2ae6f2bc92f98a
Updated order with ID 10266 with new attribute created_product_id 675ee814ae2ae6f2bc92f950
Updated order with ID 10267 with new attribute created_product_id 675ee814ae2ae6f2bc92f96c
Updated order with ID 10267 with new attribute created_product_id 675ee814ae2ae6f2bc92f97f
Updated order with ID 10267 with new attribute created_product_id 675ee814ae2ae6f2bc92f990
Updated order with ID 10268 with new attribute created_product_id 675ee814ae2ae6f2bc92f961
Updated order with ID 10268 with new attribute created_product_id 675ee814ae2ae6f2bc92f98c
Updated order with ID 10269 with new attribute created_product_id 675ee814ae2ae6f2bc92f965
```

- ✓ A.4.3 To build the link between collection categories and products, I can find the Attribute CategoryID inside products. Thus, following the same logic.

```

for product in db["products"].find():
    created_category_id = product.get("CategoryID")

    category = db["categories"].find_one({"CategoryID": created_category_id})

    db["products"].update_one(
        {"_id": product["_id"]},
        {"$set": {"created_category_id": category["_id"]}},
        upsert=False
    )

    print(f"Updated product {product['ProductName']} with new attribute created_category_id {category['_id']}")

```

```

Updated product Chai with new attribute created_category_id 675ee815ae2ae6f2bc92f992
Updated product Chang with new attribute created_category_id 675ee815ae2ae6f2bc92f992
Updated product Aniseed Syrup with new attribute created_category_id 675ee815ae2ae6f2bc92f993
Updated product Chef Anton's Cajun Seasoning with new attribute created_category_id 675ee815ae2ae6f2bc92f993
Updated product Chef Anton's Gumbo Mix with new attribute created_category_id 675ee815ae2ae6f2bc92f993
Updated product Grandma's Boysenberry Spread with new attribute created_category_id 675ee815ae2ae6f2bc92f993
Updated product Uncle Bob's Organic Dried Pears with new attribute created_category_id 675ee815ae2ae6f2bc92f998
Updated product Northwoods Cranberry Sauce with new attribute created_category_id 675ee815ae2ae6f2bc92f993
Updated product Mishi Kobe Niku with new attribute created_category_id 675ee815ae2ae6f2bc92f997
Updated product Ikura with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Queso Cabrales with new attribute created_category_id 675ee815ae2ae6f2bc92f995
Updated product Queso Manchego La Pastora with new attribute created_category_id 675ee815ae2ae6f2bc92f995
Updated product Konbu with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Tofu with new attribute created_category_id 675ee815ae2ae6f2bc92f998
Updated product Genen Shouyu with new attribute created_category_id 675ee815ae2ae6f2bc92f993
Updated product Pavlova with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Alice Mutton with new attribute created_category_id 675ee815ae2ae6f2bc92f997
Updated product Carnarvon Tigers with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Teatime Chocolate Biscuits with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Sir Rodney's Marmalade with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Sir Rodney's Scones with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Gustaf's Knäckebröd with new attribute created_category_id 675ee815ae2ae6f2bc92f996
Updated product Tunnbröd with new attribute created_category_id 675ee815ae2ae6f2bc92f996
Updated product Guaraná Fantástica with new attribute created_category_id 675ee815ae2ae6f2bc92f992
Updated product NuNuCa Nuß-Nougat-Creme with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Gumbär Gummibärchen with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Schoggi Schokolade with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Rössle Sauerkraut with new attribute created_category_id 675ee815ae2ae6f2bc92f998
Updated product Thüringer Rostbratwurst with new attribute created_category_id 675ee815ae2ae6f2bc92f997
Updated product Nord-Ost Matjeshering with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Gorgonzola Telino with new attribute created_category_id 675ee815ae2ae6f2bc92f995
Updated product Mascarpone Fabioli with new attribute created_category_id 675ee815ae2ae6f2bc92f995
Updated product Geitost with new attribute created_category_id 675ee815ae2ae6f2bc92f995
Updated product Sasquatch Ale with new attribute created_category_id 675ee815ae2ae6f2bc92f992
Updated product Steeleye Stout with new attribute created_category_id 675ee815ae2ae6f2bc92f992
Updated product Inlagd Sill with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Gravadt lax with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Côte de Blaye with new attribute created_category_id 675ee815ae2ae6f2bc92f992
Updated product Chartreuse verte with new attribute created_category_id 675ee815ae2ae6f2bc92f992
Updated product Boston Crab Meat with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Jack's New England Clam Chowder with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Singaporean Hokkien Fried Mee with new attribute created_category_id 675ee815ae2ae6f2bc92f996
Updated product Ipoh Coffee with new attribute created_category_id 675ee815ae2ae6f2bc92f992
Updated product Gula Malacca with new attribute created_category_id 675ee815ae2ae6f2bc92f993
Updated product Rogede sild with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Spegesild with new attribute created_category_id 675ee815ae2ae6f2bc92f999
Updated product Zaanse koeken with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Chocolate with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Maxilaku with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Valkoinen suklaa with new attribute created_category_id 675ee815ae2ae6f2bc92f994
Updated product Manjimup Dried Apples with new attribute created_category_id 675ee815ae2ae6f2bc92f998
Updated product Filo Mix with new attribute created_category_id 675ee815ae2ae6f2bc92f996
Updated product Perth Pasties with new attribute created_category_id 675ee815ae2ae6f2bc92f997
Updated product Tourtière with new attribute created_category_id 675ee815ae2ae6f2bc92f997
Updated product Pâté chinois with new attribute created_category_id 675ee815ae2ae6f2bc92f997
Updated product Gnocchi di nonna Alice with new attribute created_category_id 675ee815ae2ae6f2bc92f996
Updated product Ravioli Angelo with new attribute created_category_id 675ee815ae2ae6f2bc92f996
Updated product Escargots de Bourgogne with new attribute created_category_id 675ee815ae2ae6f2bc92f999

```

✓ A.4.4 Build the link between customers and orders following the same logic

```

for orders in db["orders"].find():
    created_customer_id = orders.get("CustomerID")

    customer = db["customers"].find_one({"CustomerID": created_customer_id})

```



```
print(f"Updated order with ID {order['OrderID']} with new attribute created_employee_id {employee['_id']}")
```

```
Updated order with ID 10248 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b8
Updated order with ID 10248 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b8
Updated order with ID 10248 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b8
Updated order with ID 10249 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b9
Updated order with ID 10249 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b9
Updated order with ID 10250 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10250 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10250 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10251 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b6
Updated order with ID 10251 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b6
Updated order with ID 10252 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10252 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10252 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10253 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b6
Updated order with ID 10253 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b6
Updated order with ID 10253 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b6
Updated order with ID 10254 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b8
Updated order with ID 10254 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b8
Updated order with ID 10254 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b8
Updated order with ID 10255 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bc
Updated order with ID 10255 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bc
Updated order with ID 10255 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bc
Updated order with ID 10255 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bc
Updated order with ID 10256 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b6
Updated order with ID 10256 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b6
Updated order with ID 10257 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10257 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10257 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10258 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b4
Updated order with ID 10258 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b4
Updated order with ID 10258 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b4
Updated order with ID 10259 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10259 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10260 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10260 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10260 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10261 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10261 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10262 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bb
Updated order with ID 10262 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bb
Updated order with ID 10262 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bb
Updated order with ID 10263 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bc
Updated order with ID 10263 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bc
Updated order with ID 10263 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bc
Updated order with ID 10264 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b9
Updated order with ID 10264 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b9
Updated order with ID 10265 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b5
Updated order with ID 10265 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b5
Updated order with ID 10266 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b6
Updated order with ID 10266 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10267 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10267 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b7
Updated order with ID 10268 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bb
Updated order with ID 10268 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0bb
Updated order with ID 10269 with new attribute created_employee_id 675ee80fae2ae6f2bc92f0b8
```

After building all these links, we can find that the collection **products** contains the created IDs for **supplier and category**. And the collection **orders** contain created IDs for **products, customers, and employees**. Thus, all collections are either directly, or indirectly connected to each other.

For questions B, C, and D, I will provide 2 approaches: normal approach and pipeline approach.

This is because I personally think "normal" approach is more simple, but I think the **Aggregation Pipeline** is a part of assessment for this assignment. Besides, I can compare the outputs from both approaches to verify the correctness of my work.

✓ Question (B)/(1B)

Normal Approach

For question (B), I need to list all **product names** and **unit prices** supplied by each company (supplier), along with the **supplier's name**. Because in the previous section A, i have already created a link for all collections, so I can directly call all the information I need from the **products** collection, even for the **Supplier Name information**.

```
Cursor_Object = db["products"].find({}, {"ProductName": 1, "UnitPrice": 1, "created_supplier_id": 1})
# Cursor Object is an iterable object, so I can loop through it to get the data:
for product in Cursor_Object:
    supplier = db["suppliers"].find_one({"_id": product["created_supplier_id"]}, {"CompanyName": 1})
    print({
        "SupplierName": supplier["CompanyName"],
        "ProductName": product["ProductName"],
        "UnitPrice": product["UnitPrice"]
    })
```

```
{
  "SupplierName": "Specialty Biscuits, Ltd.",
  "ProductName": "Chai",
  "UnitPrice": 18.0
}
{
  "SupplierName": "Exotic Liquids",
  "ProductName": "Chang",
  "UnitPrice": 19.0
}
{
  "SupplierName": "Exotic Liquids",
  "ProductName": "Aniseed Syrup",
  "UnitPrice": 10.0
}
{
  "SupplierName": "New Orleans Cajun Delights",
  "ProductName": "Chef Anton's Cajun Seasoning",
  "UnitPrice": 22.0
}
{
  "SupplierName": "New Orleans Cajun Delights",
  "ProductName": "Chef Anton's Gumbo Mix",
  "UnitPrice": 21.35
}
{
  "SupplierName": "Grandma Kelly's Homestead",
  "ProductName": "Grandma's Boysenberry Spread",
  "UnitPrice": 25.0
}
{
  "SupplierName": "Grandma Kelly's Homestead",
  "ProductName": "Uncle Bob's Organic Dried Pears",
  "UnitPrice": 30.0
}
{
  "SupplierName": "Grandma Kelly's Homestead",
  "ProductName": "Northwoods Cranberry Sauce",
  "UnitPrice": 40.0
}
{
  "SupplierName": "Tokyo Traders",
  "ProductName": "Mishi Kobe Niku",
  "UnitPrice": 97.0
}
{
  "SupplierName": "Tokyo Traders",
  "ProductName": "Ikura",
  "UnitPrice": 31.0
}
{
  "SupplierName": "Cooperativa de Quesos 'Las Cabras'",
  "ProductName": "Queso Cabrales",
  "UnitPrice": 21.0
}
{
  "SupplierName": "Cooperativa de Quesos 'Las Cabras'",
  "ProductName": "Queso Manchego La Pastora",
  "UnitPrice": 38.0
}
{
  "SupplierName": "Mayumi's",
  "ProductName": "Konbu",
  "UnitPrice": 6.0
}
{
  "SupplierName": "Mayumi's",
  "ProductName": "Tofu",
  "UnitPrice": 23.25
}
{
  "SupplierName": "Mayumi's",
  "ProductName": "Genen Shouyu",
  "UnitPrice": 13.0
}
{
  "SupplierName": "Pavlova, Ltd.",
  "ProductName": "Pavlova",
  "UnitPrice": 17.45
}
{
  "SupplierName": "Pavlova, Ltd.",
  "ProductName": "Alice Mutton",
  "UnitPrice": 39.0
}
{
  "SupplierName": "Pavlova, Ltd.",
  "ProductName": "Carnarvon Tigers",
  "UnitPrice": 62.5
}
{
  "SupplierName": "Specialty Biscuits, Ltd.",
  "ProductName": "Teatime Chocolate Biscuits",
  "UnitPrice": 9.2
}
{
  "SupplierName": "Specialty Biscuits, Ltd.",
  "ProductName": "Sir Rodney's Marmalade",
  "UnitPrice": 81.0
}
{
  "SupplierName": "Specialty Biscuits, Ltd.",
  "ProductName": "Sir Rodney's Scones",
  "UnitPrice": 10.0
}
{
  "SupplierName": "PB Knäckebröd AB",
  "ProductName": "Gustaf's Knäckebröd",
  "UnitPrice": 21.0
}
{
  "SupplierName": "PB Knäckebröd AB",
  "ProductName": "Tunnbröd",
  "UnitPrice": 9.0
}
{
  "SupplierName": "Refrescos Americanas LTDA",
  "ProductName": "Guaraná Fantástica",
  "UnitPrice": 4.5
}
{
  "SupplierName": "Heli Süßwaren GmbH & Co. KG",
  "ProductName": "NuNuCa Nuß-Nougat-Creme",
  "UnitPrice": 14.0
}
{
  "SupplierName": "Heli Süßwaren GmbH & Co. KG",
  "ProductName": "Gumbär Gummibärchen",
  "UnitPrice": 31.23
}
{
  "SupplierName": "Heli Süßwaren GmbH & Co. KG",
  "ProductName": "Schoggi Schokolade",
  "UnitPrice": 43.9
}
{
  "SupplierName": "Plutzer Lebensmittelgroßmärkte AG",
  "ProductName": "Rössle Sauerkraut",
  "UnitPrice": 45.6
}
{
  "SupplierName": "Plutzer Lebensmittelgroßmärkte AG",
  "ProductName": "Thüringer Rostbratwurst",
  "UnitPrice": 123.79
}
{
  "SupplierName": "Nord-Ost-Fisch Handelsgesellschaft mbH",
  "ProductName": "Nord-Ost Matjeshering",
  "UnitPrice": 25.89
}
{
  "SupplierName": "Formaggi Fortini s.r.l.",
  "ProductName": "Gorgonzola Telino",
  "UnitPrice": 12.5
}
{
  "SupplierName": "Formaggi Fortini s.r.l.",
  "ProductName": "Mascarpone Fabioli",
  "UnitPrice": 32.0
}
{
  "SupplierName": "Norske Meierier",
  "ProductName": "Geitost",
  "UnitPrice": 2.5
}
{
  "SupplierName": "Bigfoot Breweries",
  "ProductName": "Sasquatch Ale",
  "UnitPrice": 14.0
}
{
  "SupplierName": "Bigfoot Breweries",
  "ProductName": "Steeleye Stout",
  "UnitPrice": 18.0
}
{
  "SupplierName": "Svensk Sjöföda AB",
  "ProductName": "Inlagd Sill",
  "UnitPrice": 19.0
}
{
  "SupplierName": "Svensk Sjöföda AB",
  "ProductName": "Gravad lax",
  "UnitPrice": 26.0
}
{
  "SupplierName": "Aux joyeux ecclésiastiques",
  "ProductName": "Côte de Blaye",
  "UnitPrice": 263.5
}
{
  "SupplierName": "Aux joyeux ecclésiastiques",
  "ProductName": "Chartreuse verte",
  "UnitPrice": 18.0
}
{
  "SupplierName": "New England Seafood Cannery",
  "ProductName": "Boston Crab Meat",
  "UnitPrice": 18.4
}
{
  "SupplierName": "New England Seafood Cannery",
  "ProductName": "Jack's New England Clam Chowder",
  "UnitPrice": 9.65
}
{
  "SupplierName": "Leka Trading",
  "ProductName": "Singaporean Hokkien Fried Mee",
  "UnitPrice": 14.0
}
{
  "SupplierName": "Leka Trading",
  "ProductName": "Ipoh Coffee",
  "UnitPrice": 46.0
}
{
  "SupplierName": "Leka Trading",
  "ProductName": "Gula Malacca",
  "UnitPrice": 19.45
}
{
  "SupplierName": "Lyngbysild",
  "ProductName": "Rogede sild",
  "UnitPrice": 9.5
}
{
  "SupplierName": "Lyngbysild",
  "ProductName": "Spegesild",
  "UnitPrice": 12.0
}
{
  "SupplierName": "Zaanse Snoepfabriek",
  "ProductName": "Zaanse koeken",
  "UnitPrice": 9.5
}
{
  "SupplierName": "Zaanse Snoepfabriek",
  "ProductName": "Chocolade",
  "UnitPrice": 12.75
}
{
  "SupplierName": "Karkki Oy",
  "ProductName": "Maxilaku",
  "UnitPrice": 20.0
}
{
  "SupplierName": "Karkki Oy",
  "ProductName": "Valkoinen suklaa",
  "UnitPrice": 16.25
}
{
  "SupplierName": "G'day, Mate",
  "ProductName": "Manjimup Dried Apples",
  "UnitPrice": 53.0
}
{
  "SupplierName": "G'day, Mate",
  "ProductName": "Filo Mix",
  "UnitPrice": 7.0
}
{
  "SupplierName": "G'day, Mate",
  "ProductName": "Perth Pasties",
  "UnitPrice": 32.8
}
{
  "SupplierName": "Ma Maison",
  "ProductName": "Tourtière",
  "UnitPrice": 7.45
}
{
  "SupplierName": "Ma Maison",
  "ProductName": "Pâté chinois",
  "UnitPrice": 24.0
}
{
  "SupplierName": "Pasta Buttini s.r.l.",
  "ProductName": "Gnocchi di nonna Alice",
  "UnitPrice": 38.0
}
{
  "SupplierName": "Pasta Buttini s.r.l.",
  "ProductName": "Ravioli Angelo",
  "UnitPrice": 19.5
}
{
  "SupplierName": "Escargots Nouveaux",
  "ProductName": "Escargots de Bourgogne",
  "UnitPrice": 13.25
}
```

▼ Pipeline Approach

use `lookup` * `to join the suppliers collection with the products collection`. Then use * `project` select information I need, in this question, they are `CompanyName`, `SupplierName`, and `UnitPrice`.

```
pipeline_question_b = [
    {
        "$lookup": {
            "from": "products",
            "localField": "SupplierID",
            "foreignField": "SupplierID",
            "as": "supplier_products"
        }
    },
    {
        "$project": {
            "SupplierName": "$CompanyName",
            "Products": {
                "$map": {
                    "input": "$supplier_products",
                    "as": "product",
                    "in": {
                        "ProductName": "$$product.ProductName",
                        "UnitPrice": "$$product.UnitPrice"
                    }
                }
            }
        }
    }
]

results_b = db["suppliers"].aggregate(pipeline_question_b)

print("Products Supplied by Each Company:")
for result in results_b:
    print(f"Supplier Name: {result['SupplierName']}")
    print("Products:")
    for product in result["Products"]:
        print(f"    {product['ProductName']} (Unit Price: {product['UnitPrice']})")
# separate into different sections, for better readability.
print("-" * 50)
```

```
➦ Products Supplied by Each Company:
Supplier Name: Exotic Liquids
Products:
- Chang (Unit Price: 19.0)
- Aniseed Syrup (Unit Price: 10.0)
-----
Supplier Name: New Orleans Cajun Delights
Products:
- Chef Anton's Cajun Seasoning (Unit Price: 22.0)
- Chef Anton's Gumbo Mix (Unit Price: 21.35)
- Louisiana Fiery Hot Pepper Sauce (Unit Price: 21.05)
- Louisiana Hot Spiced Okra (Unit Price: 17.0)
-----
Supplier Name: Grandma Kelly's Homestead
Products:
- Grandma's Boysenberry Spread (Unit Price: 25.0)
- Uncle Bob's Organic Dried Pears (Unit Price: 30.0)
- Northwoods Cranberry Sauce (Unit Price: 40.0)
-----
Supplier Name: Tokyo Traders
Products:
- Mishi Kobe Niku (Unit Price: 97.0)
- Ikura (Unit Price: 31.0)
- Longlife Tofu (Unit Price: 10.0)
-----
Supplier Name: Cooperativa de Quesos 'Las Cabras'
Products:
- Queso Cabrales (Unit Price: 21.0)
- Queso Manchego La Pastora (Unit Price: 38.0)
-----
Supplier Name: Mayumi's
Products:
- Konbu (Unit Price: 6.0)
- Tofu (Unit Price: 23.25)
- Genen Shouyu (Unit Price: 13.0)
```

```

-----
Supplier Name: Pavlova, Ltd.
Products:
- Pavlova (Unit Price: 17.45)
- Alice Mutton (Unit Price: 39.0)
- Carnarvon Tigers (Unit Price: 62.5)
- Vegie-spread (Unit Price: 43.9)
- Outback Lager (Unit Price: 15.0)
-----

Supplier Name: Specialty Biscuits, Ltd.
Products:
- Chai (Unit Price: 18.0)
- Teatime Chocolate Biscuits (Unit Price: 9.2)
- Sir Rodney's Marmalade (Unit Price: 81.0)
- Sir Rodney's Scones (Unit Price: 10.0)
- Scottish Longbreads (Unit Price: 12.5)
-----

Supplier Name: PB Knäckebröd AB
Products:
- Gustaf's Knäckebröd (Unit Price: 21.0)
- Tunnbröd (Unit Price: 9.0)
-----

```

Supplier Name: PB Knäckebröd AB

▼ Question (C)/(1C)

Normal Approach

In question C, I need to list the top 10 best-seller products'categories. In other words, I will rank products based on their aggregate sales, then pick the first 10 from the ranking list. Find these first 10 products' categories.

I am a bit confused about the term "10-best-seller", I will understand it as the top 10 products that has been purchased most frequently, instead of the sales/revenues.

Holding this logic, I will break down codes into several parts

Part 1: find the aggregate sales for each product by looking into information from **orders**.

Part 2: rank them, and get the first 10.

Part 3: get the category names for those first 10.

Part 4: interpretation.

The code cell is a bit long, so inside code cells, I will label which part is the code below processing on.

```

# Part 1
# create an empty dictionary.
product_sales_dict = {}

# specify calculation formula.
for order in db["orders"].find():
    product_id = order.get("ProductID")
    quantity = order.get("Quantity", 0)

# check if product_id is in the dictionary, if true , add total to the existing value, if false, create a new key-value pair.
if product_id in product_sales_dict:
    product_sales_dict[product_id] += quantity
else:
    product_sales_dict[product_id] = quantity

# Part 2
# top_products stores products's id and total sale numbers, top_product_ids only stores id.
top_products = sorted(product_sales_dict.items(), key=lambda x: x[1], reverse=True)[:10]
top_product_ids = [product_id for product_id, _ in top_products]

# Part 3
# create an empty category list.
categories = []

# Locate these 10 products, and get their full information, not just id this time.
for product_id in top_product_ids:
    product = db["products"].find_one({"ProductID": product_id})

# get category information, then append them to the category list.
if product and "created_category_id" in product:
    category = db["categories"].find_one({"_id": product["created_category_id"]})
    categories.append({"CategoryName": category["CategoryName"]})

```

```
# Part 4
for category in categories:
    print(category)
print(f'The order from top to bottom is sales descending from No.1 "product\'s category" to No.10 "product\'s category".')
```

⇒

```
{'CategoryName': 'Dairy Products'}
{'CategoryName': 'Dairy Products'}
{'CategoryName': 'Dairy Products'}
{'CategoryName': 'Grains/Cereals'}
{'CategoryName': 'Confections'}
{'CategoryName': 'Beverages'}
{'CategoryName': 'Beverages'}
{'CategoryName': 'Seafood'}
{'CategoryName': 'Confections'}
{'CategoryName': 'Beverages'}
The order from top to bottom is sales descending from No.1 "product's category" to No.10 "product's category".
```

✓ Pipeline Approach

Overall the query language is easy to understand for this question C, so I will just explain some special part of this query:

"**sort**": "**TotalQuantity**": -1 * *means I order the category in a descending order based on quantity sold.* ** "**limit**": 10 means I only need the top 10 categories.

"**\$addFields**" The purpose for creating a new field is because I want to store the value of total quantity sold for every different product into a new array. Here it is called TotalQuantity.

```
pipeline_question_c = [
    {
        "$lookup": {
            "from": "orders",
            "localField": "_id",
            "foreignField": "created_product_id",
            "as": "product_orders"
        }
    },
    {
        "$addFields": {
            "TotalQuantity": {"$sum": "$product_orders.Quantity"}
        }
    },
    {
        "$sort": {"TotalQuantity": -1}
    },
    {
        "$limit": 10
    },
    {
        "$lookup": {
            "from": "categories",
            "localField": "created_category_id",
            "foreignField": "_id",
            "as": "category"
        }
    },
    {
        "$unwind": "$category"
    },
    {
        "$project": {
            "CategoryName": "$category.CategoryName"
        }
    }
]
```

```
results_c = db["products"].aggregate(pipeline_question_c)

print("Categories for Top 10 Selling Products are:")
for result in results_c:
    print(f"Category Name: {result['CategoryName']}")
```

⇒ Categories for Top 10 Selling Products are:
 Category Name: Dairy Products
 Category Name: Dairy Products

```

Category Name: Dairy Products
Category Name: Grains/Cereals
Category Name: Confections
Category Name: Beverages
Category Name: Beverages
Category Name: Seafood
Category Name: Confections
Category Name: Dairy Products

```

This is the advantage of using 2 approaches, For question C, I can see that the outputs are same between approaches, so I can be certain that the answer is correct.

✓ Question (D)/(1D)

In Question (D), I need to list all products purchased by every customer.

In Question (A), I have already linked collections. In collection **orders**, I can find the created ids for **Product** and **Customer**. Therefore, I can use orders as bridge to get everything I need.

To tackle this question, I have gained some insights from the question C: using dictionary and list to store the output, I will create a dictionary first, then store every information I need like "**Company/Contact Names**" and "**Product Names**" into lists, which could be included into the dictionary.

Note: Because the question did not mention which name should be referred as customer name, I include both the **Company Name** and **Contact Name** in the result.

```

customer_products = {}

for customer in db["customers"].find():
    customer_company_name = customer["CompanyName"]
    customer_contact_name = customer["ContactName"]
    customer_products[customer_company_name] = []

    orders = db["orders"].find({"created_customer_id": customer["_id"]})
    for _ in orders:
        product_id = _.get("created_product_id")
        product = db["products"].find_one({"_id": product_id})
        customer_products[customer_company_name].append(product["ProductName"])

# Print Customer Company Name, Customer Contact Name, and Products Bought for each Customer:
print(f"Customer Company Name: {customer_company_name}")
print(f"Customer Contact Name: {customer_contact_name}")
print("Products Bought:")
for product in customer_products[customer_company_name]:
    # use "-" at the front for better readability, since there are usually many products bought by each customer.
    print(f"- {product}")

```

```

🔗 Customer Company Name: Alfreds Futterkiste
Customer Contact Name: Maria Anders
Products Bought:
- Rössle Sauerkraut
- Chartreuse verte
- Spegesild
- Vegie-spread
- Aniseed Syrup
- Lakkalikööri
- Raclette Courdavault
- Original Frankfurter grüne Soße
- Grandma's Boysenberry Spread
- Rössle Sauerkraut
- Escargots de Bourgogne
- Flotemysost
Customer Company Name: Ana Trujillo Emparedados y helados
Customer Contact Name: Ana Trujillo
Products Bought:
- Gudbrandsdalsost
- Outback Lager
- Tofu
- Singaporean Hokkien Fried Mee
- Camembert Pierrot
- Mascarpone Fabioli
- Queso Cabrales

```

- Konbu
- Teatime Chocolate Biscuits
- Mozzarella di Giovanni

Customer Company Name: Antonio Moreno Taqueria
Customer Contact Name: Antonio Moreno

Products Bought:

- Queso Cabrales
- Ipoh Coffee
- Chocolate
- Queso Cabrales
- Boston Crab Meat
- Ravioli Angelo
- Raclette Courdavault
- Alice Mutton
- Sasquatch Ale
- Perth Pasties
- Gumbär Gummibärchen
- Geitost
- Geitost
- Louisiana Hot Spiced Okra
- Rhönbräu Klosterbier
- Chang
- Singaporean Hokkien Fried Mee

Customer Company Name: Around the Horn
Customer Contact Name: Thomas Hardy

Products Bought:

- Guaraná Fantástica
- Ravioli Angelo
- Konbu
- Valkoinen suklaa
- Gnocchi di nonna Alice
- Chocolate
- Outback Lager

▼ Pipeline Approach

The pipeline approach follow the same logic as normal approach, I join tables with *lookup ***, *thenIbreakarrayswith **unwind*. This

- ▼ is crucial because otherwise products bought will be presented as a very long list (because each product is not treated separately), which is unfriendly for reading. Finally, I get information I need with **\$group**.

```
pipeline_question_d = [
  {
    "$lookup": {
      "from": "orders",
      "localField": "_id",
      "foreignField": "created_customer_id",
      "as": "customer_orders"
    }
  },
  {
    "$unwind": {
      "path": "$customer_orders",
      "preserveNullAndEmptyArrays": True
    }
  },
  {
    "$lookup": {
      "from": "products",
      "localField": "customer_orders.created_product_id",
      "foreignField": "_id",
      "as": "ordered_products"
    }
  },
  {
    "$unwind": {
      "path": "$ordered_products",
      "preserveNullAndEmptyArrays": True
    }
  },
  {
    "$group": {
      "_id": {
        "CustomerCompanyName": "$CompanyName",
        "CustomerContactName": "$ContactName"
      },
      "ProductsBought": {"$addToSet": "$ordered_products.ProductName"}
    }
  }
]
```

```

    }
}

]

results_d = db["customers"].aggregate(pipeline_question_d)

print("Products Bought by Each Customer:")
for result in results_d:
    print(f"Customer Company Name: {result['_id']['CustomerCompanyName']}")
    print(f"Customer Contact Name: {result['_id']['CustomerContactName']}")
    print("Products Bought:")
    for product in result["ProductsBought"]:
        print(f"- {product}")

```

```

➡ Products Bought by Each Customer:
Customer Company Name: Richter Supermarkt
Customer Contact Name: Michael Holz
Products Bought:
- Konbu
- Rhönbräu Klosterbier
- Spegesild
- Ravioli Angelo
- Geitost
- Lakkalikööri
- Outback Lager
- Escargots de Bourgogne
- Mozzarella di Giovanni
- Camembert Pierrot
- Jack's New England Clam Chowder
- Perth Pasties
- Louisiana Fiery Hot Pepper Sauce
- Thüringer Rostbratwurst
- Gudbrandsdalsost
- Manjimup Dried Apples
- Inlagd Sill
- Valkoinen suklaa
- Röd Kaviar
- Chang
- Gorgonzola Telino
- Nord-Ost Matjeshering
- Raclette Courdavault
- Gumbär Gummibärchen
- Pavlova
- Filo Mix
Customer Company Name: Victuailles en stock
Customer Contact Name: Mary Saveley
Products Bought:
- Jack's New England Clam Chowder
- NuNuCa Nuß-Nougat-Creme
- Manjimup Dried Apples
- Gustaf's Knäckebröd
- Ikura
- Louisiana Fiery Hot Pepper Sauce
- Ravioli Angelo
- Tarte au sucre
- Sirop d'érable
- Geitost
- Mozzarella di Giovanni
- Outback Lager
- Scottish Longbreads
- Filo Mix
- Chang
- Spegesild
- Steeleye Stout
- Uncle Bob's Organic Dried Pears
- Tourtière
- Ipoh Coffee
- Longlife Tofu
- Chocolate
Customer Company Name: Ottilies Käseladen
Customer Contact Name: Henriette Pfalzheim
Products Bought:

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